Can published ARDS subphenotype classifiers survive routine ICU data?

A reproduction and deployment stress-test across six open datasets

James Conde

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Abstract

Background. The hyperinflammatory and hypoinflammatory subphenotypes of the acute respiratory distress syndrome are established and independently replicated, and have been reduced to parsimonious classifiers intended for bedside use. Every published validation of those classifiers, however, has run in curated research cohorts, where variables were measured on protocol and missingness was low by design. Whether they survive routine care is untested.

Methods. Six open datasets, three modalities. A reference two-class assignment was derived by Gaussian mixture on 4,680 patients from one hospital system of the PhysioNet/CinC Challenge 2019 corpus, then reproduced by classifiers restricted to per-patient values and verified portable by two independent checks. The frozen models were applied to a second hospital system, and to the eICU demonstration database under an entirely different schema. A modified SOFA score, a variable-ablation curve, a prognostic enrichment simulation and one iterative refinement cycle were pre-specified. A molecular layer used three transcriptomic cohorts and a 1,472-assay proteomic cohort.

Results. The expected two-class directional pattern was recovered at 28.5 % prevalence, and portable classifiers reproduced that reference assignment ($\kappa = 0.852$). Transfer was limited not by the model but by measurement: the frozen classifier scored 82.4 % of second-hospital patients using coverage-matched variables and 1.6 % using the variables that performed best, because bicarbonate is charted for 97.4 % of patients at one site and 2.1 % at the other. Expressed as trial screening burden, this is a 52-fold difference. Against sepsis onset the classes appeared near-worthless, adding +0.017 AUC over a severity score; against mortality in eICU the same frozen model separated 16.8 % from 36.6 % hospital mortality ($p < 0.0001$).

Conclusion. Reproducibility, portability and deployability are distinct properties. A classifier can recover a published structure, run without cohort-derived preprocessing, and still be unusable at another hospital because the measurements it requires are not routinely obtained. The apparent weakness of the classes on the primary cohort depended strongly on the available endpoint.

1 Introduction

Latent class analysis of two randomised trials identified two inflammatory subphenotypes of ARDS [1], a finding since recovered across five or more cohorts and shown to carry differential mortality [2] and differential treatment response. Subsequent work reduced the classification to parsimonious models usable from routinely available data [3, 4], and validated those models in observational cohorts [5]. **This study reproduces that established structure; it does not discover it.** Any reading of what follows as a novel finding about subphenotype existence would be a misreading.

What remains open is deployability. The validations above ran in ARDSNet trials, EARLI, VALID and LUNG SAFE — cohorts in which the classifier's input variables were measured on protocol. Routine intensive care is not like that. Laboratory measurement is intermittent, ordering behaviour differs between institutions, and missingness in electronic health records is well established as informative rather than random, reflecting care process rather than patient biology alone.

Three properties are conflated in most of this literature, and this study separates them. *Reproducibility* is whether a published structure can be recovered in a new dataset. *Portability* is whether a model can run without cohort-derived preprocessing — no z-scores against a population, no median imputation. *Deployability* is whether the measurements the model requires are actually obtained at the site where it would run. A classifier can satisfy the first two completely and fail the third, and the third is where this study's contribution lies.

This gives a concrete operational question, and it is the question a clinical-development team faces when designing an enrichment strategy: *do published subphenotype classifiers survive routine, sparsely measured, unadjudicated ICU data, and how gracefully do they degrade as the variables they need go missing?*

The study is deliberately constrained to open-access data. That costs real capability — most consequentially, the primary cohort has no mortality outcome — and buys schedule certainty and full reproducibility.

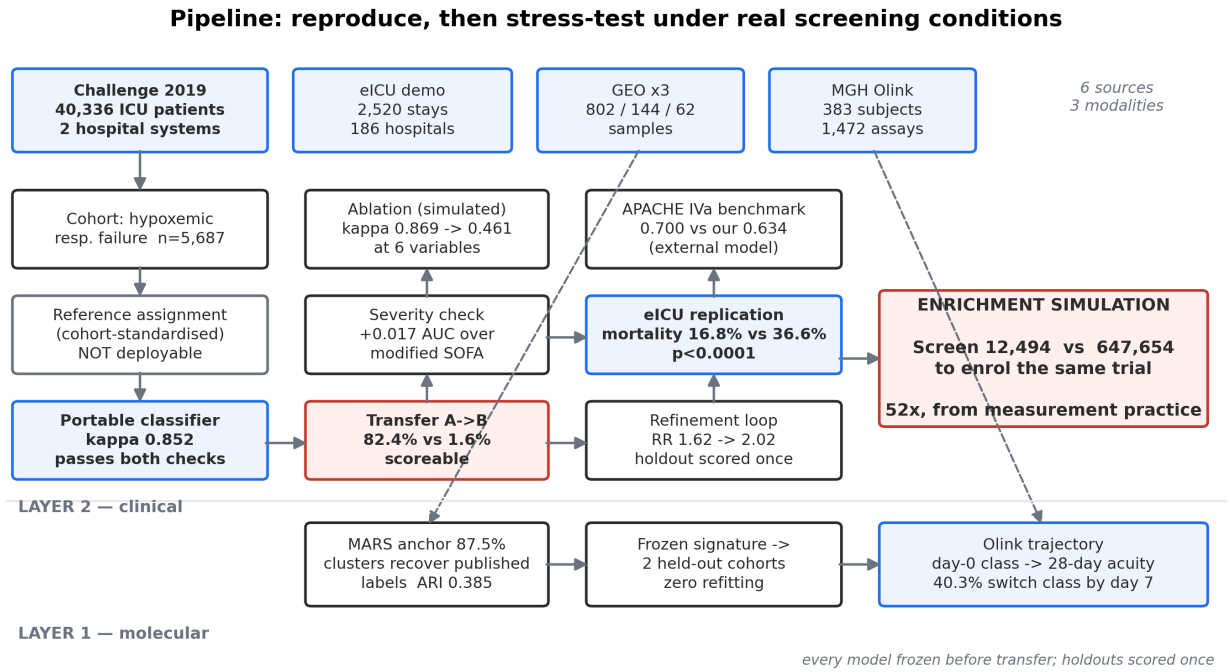


Figure 1. Analysis pipeline. Six sources, three modalities. Every model is frozen before it crosses a dataset boundary, and the refinement loop's holdout is scored exactly once

2 Methods

2.1 Data sources

Six open-access sources were used (Table 1). Attribution requirements are met in `CITATIONS.md`.

Source	Size	Role
PhysioNet/CinC Challenge 2019	40,336 patients, 2 systems	Primary cohort
eICU-CRD Demo	2,520 unit stays	Mortality; external benchmark
MIMIC-IV Demo	100 patients	Schema familiarity, smoke tests
GSE65682 (MARS)	802 samples, 479 labelled	Transcriptomic discovery, anchor
GSE32707, GSE66890	144 and 62 samples	Transcriptomic transfer
MGH Olink Explore 1536	383 subjects, 1,472 assays	Proteomics, longitudinal

Table 1. Data sources

2.2 Cohort definition

Berlin criteria cannot be approximated on this data path: there is no imaging, no PEEP and no ventilation flag. The cohort is therefore **hypoxemic respiratory failure identified by an SpO_2/FiO_2 surrogate**, and is never described as ARDS. A pre-commit guard enforces that wording mechanically.

Inclusion required age ≥ 18, at least 24 hours of record, and at least one hour with $S/F \leq 235$ (a validated surrogate for $P/F \leq 200$). Hours with $SpO_2 \geq 97$ were excluded because the ratio is uninformative once the oximeter saturates, as were qualifying hours beyond ICU hour 168. All thresholds were frozen before any data was read.

Observed FiO_2 ranged from 0.000 to 10.000 on a fraction scale where room air is 0.209; 321 of 112,155 values (0.29 %) are physiologically impossible. A validity window of $[0.21, 1.0]$ was pre-registered. The errors are asymmetric — a value above 1.0 *deflates* S/F and would falsely admit a patient as severely hypoxemic — which is why they are filtered explicitly rather than dismissed as rare.

2.3 Features and the treatment of missingness

Static features summarise T0 to T0+24 h; longitudinal features bin T0 to T0+72 h into twelve 6-hour intervals. **Every value channel carries a parallel binary mask** recording whether the measurement was taken, because forward-filling without one fabricates trajectories. Nothing is imputed at any point: cohort-median imputation would introduce a population statistic into the inference path, which the portability checks of Section 2.5 exist to forbid.

A third channel, `in_record`, marks whether the patient was still in the ICU during each bin. Without it the mask conflates *not ordered* with *already discharged*. The distinction is large rather than pedantic: overall observation density is 27.2 % raw but 52.5 % within the record, so nearly half of apparent missingness is censoring.

2.4 Reference assignment, and what "reproduction" means here

Four distinct objects are involved, and conflating them would overstate the result: (1) the **published** subphenotype structure, derived by latent class analysis on trial cohorts [1–3]; (2) a **reference assignment** fitted here, on this cohort, by Gaussian mixture; (3) a **portable approximation** of that reference assignment; and (4) **external prognostic validation** of the portable model in an independent database.

This study demonstrates that (2) recovers the *directional pattern* of (1), that (3) reproduces (2) at high agreement, and that (4) holds. It does **not** demonstrate that (2) is the same partition as the published latent classes: published class assignments for these patients do not exist, so no direct

agreement can be computed. Where the text says "recovered" it means the directional pattern; "reproduced" means agreement with the reference assignment fitted here.

A Gaussian mixture on log-transformed, **cohort-standardised** static features, fitted on hospital system A. It requires the cohort mean and standard deviation of every variable at inference and is therefore not deployable by construction. It exists to generate labels, not to ship. k was selected by BIC over a pre-registered range of 2–4 with 100 restarts.

2.5 Portable classifiers and how portability was verified

Three candidate forms — logistic regression on raw values, a depth-4 decision tree with absolute cut-points, and gradient boosting — were trained to reproduce the reference labels using only per-patient values. The portability claim is made **executable rather than asserted**, by two independent checks:

1. *Static introspection* walks the fitted estimator and rejects any component carrying a cohort-derived statistic applied at transform time — scalers, imputers, PCA.
2. *Functional batch-invariance* predicts each patient alone and again inside the full batch and requires the two to be identical. This is the check that binds, because a model can pass introspection and still normalise inside a custom predict function.

2.6 Cross-hospital transfer

Models fitted on system A were frozen and applied to system B with no refitting, under two feature sets whose contrast isolates the mechanism: a **restricted** set of 14 variables whose coverage differs by less than 10 percentage points between sites, and a **naive** set of 12 including the blood-gas block. Because system B contains no patients with reference labels, transfer is evaluated by scoreability, predicted prevalence, and whether the predicted class still stratifies sepsis onset — an endpoint recorded independently of whether the classifier can run.

2.7 Severity comparison

Severity is a **modified** SOFA, and must not be called SOFA. Two of six organ systems are unavailable: cardiovascular is capped at 1 point for want of vasopressor dosing, and CNS is omitted entirely for want of GCS. Maximum attainable score is 17 rather than 24, and bilirubin is missing for 77 % of patients so the liver component scores zero by default for most. Every one of these deficiencies weakens the severity score and therefore **biases the comparison in the endotype's favour**.

2.8 Ablation

Three analyses. **(A)** Progressive removal of variables, ordered least-available-first rather than least-important-first, because a deployment loses whatever the site does not measure. **(B)** Natively sparse versus natively complete patients. **(C)** A model trained on the missingness mask alone, with every value discarded.

Analyses B and C run on the full cohort against sepsis onset rather than on labelled patients. This is not a convenience: endotype labels exist only for the reference model's complete cases, within which the mask is all-ones by construction, so labelled patients cannot support any question *about* missingness (Section 6).

2.9 Enrichment simulation

Prognostic enrichment only — a power calculation, not a causal claim. Given observed event rates per class, required sample size was computed for unstratified enrollment and for enrollment restricted to the higher-event-rate class, at 80 % power, $\alpha = 0.05$ and an assumed 20 % relative risk reduction. Misclassification was modelled through positive predictive value, and screening burden from the fraction of patients who can be scored at all. **Predictive** enrichment — which patients benefit more from treatment — is a different question requiring a treatment-effect estimate, is confounded by indication in observational data, and was not attempted.

2.10 Refinement loop

One documented cycle. System B was split once with a fixed seed into a diagnosis slice and a final holdout. Round 1 predicted on the diagnosis slice; errors were stratified by missingness, age and predicted class into a taxonomy; refinements were selected by **rules written down before any diagnosis ran**; round 2 was evaluated on the holdout, scored exactly once.

2.11 Molecular layer (secondary)

This layer is biological triangulation, not validation of the clinical classifier. The two cannot validate one another: the cohorts share no patients, the modalities differ, and no patient-level link exists. It is included because it tests whether an inflammatory two-class structure is recoverable in molecular data at all, and because the longitudinal proteomics answer a question the clinical layer cannot — whether class membership is stable over time.

A regularised multinomial model was trained on GSE65682 to predict published MARS endotype labels, with the resulting signature frozen. Consensus clustering with PAC-based k selection was run independently. The frozen signature was scored on two held-out cohorts on different array platforms with zero refitting; **cohorts were never merged**, since merging across platforms manufactures clusters that separate by batch. Olink day-0 NPX was clustered and related to a five-level acuity ordinal at days 0, 3, 7 and 28. Pathway sets were pre-registered before any clustering ran.

2.12 Pre-registration, guards and reproducibility

Thresholds, feature sets and gene sets were committed before the analyses that use them, so commit ordering makes the freeze auditable. Every extraction step carries an assertion checkpoint with an expected row-count range. Pre-commit guards block row-level data, discovery language, the ARDS-vocabulary violation and credential leaks. Thirty-three tests cover the guards and the consensus-clustering implementation.

3 Results

3.1 Cohort

From 40,336 patients, 5,687 met criteria (4,680 in system A, 1,007 in system B; Table 7). The cohort is selected rather than representative: requiring a recorded FiO_2 picks out patients who are older, stay longer and have roughly triple the sepsis rate, with standardised differences of 0.19 to 0.34. System B's cohort has a higher baseline sepsis rate than system A's (22.2 % versus 13.9 %).

3.2 The expected directional pattern is recovered

At $k = 2$, the class with the lowest bicarbonate showed **all four directions the published literature reports** for the hyperinflammatory class — lower bicarbonate, higher creatinine, higher white cell count, higher event rate — at a prevalence of 28.5 %, inside the 20–35 % range published cohorts report. These four were specified in advance as the directional signature to check; agreement with the published *assignment* is not computable, since published labels for these patients do not exist.

BIC preferred $k = 4$. That value is the maximum of the pre-registered range with BIC still falling, making it a **boundary solution**: the range, not the criterion, chose k. The range was not widened afterwards, since doing so would be selecting k by result.

3.3 Portable classifiers reproduce the reference assignment

Model	κ	AUC	Brier
Gradient boosting	0.852	0.987	0.041
Logistic regression	0.723	0.932	0.082
Decision tree (depth 4)	0.671	0.871	0.093

Table 2. Agreement with the reference assignment, out-of-fold. All three models passed both portability checks

The tree ships only split thresholds and is readable as a bedside rule. The parsimony curve reaches its ceiling at four variables: BUN, bicarbonate, creatinine and pH. **Two of those four are not recorded comparably at the second hospital.**

3.4 Transfer is limited by measurement, not by the model

Feature set	Variables	Scoreable, system A	Scoreable, system B
Restricted (coverage-matched)	14	79.8 %	**82.4 %**
Naive (includes blood gases)	12	70.0 %	**1.6 %**

Table 3. Fraction of the cohort a frozen classifier can score at each site

The gap between 82.4 % and 1.6 % is measurement practice, isolated cleanly because the restricted set's coverage is matched across sites by construction. Bicarbonate is recorded for 97.4 % of system A patients and 2.1 % of system B; chloride 97.3 % versus 8.1 %; base excess 62.2 % versus 2.8 %. The difference runs in both directions — system B charts diastolic pressure (99.9 % versus 63.7 %) and troponin (32.8 % versus 2.4 %) *more* (Figure 2) — which identifies it as institutional measurement behaviour rather than a data-quality defect at one site.

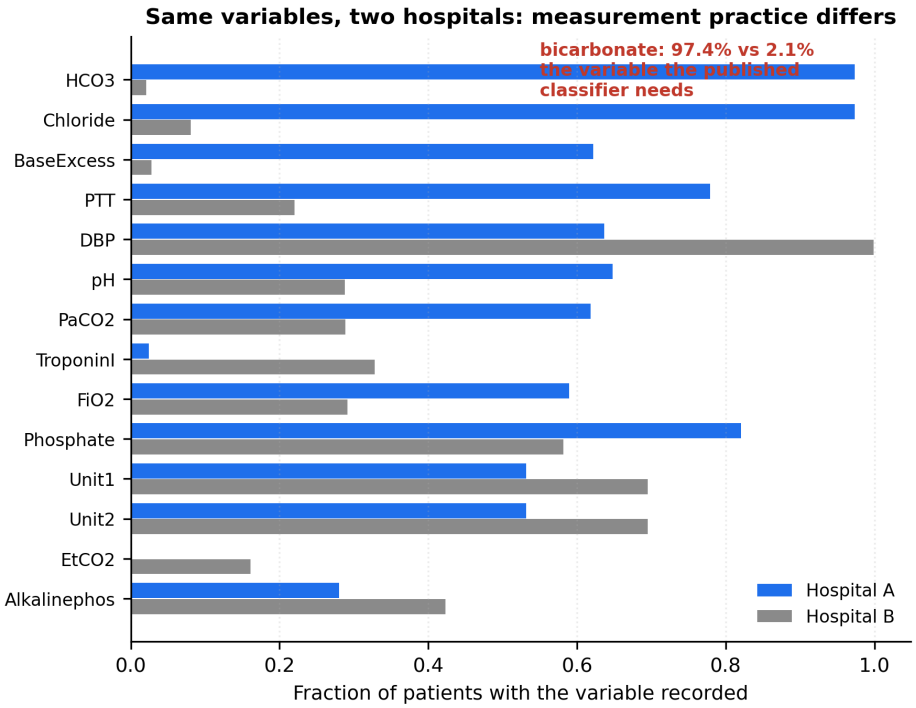


Figure 2. Patient-level coverage for the variables differing most between the two hospital systems. The difference runs in both directions, which is why this is measurement practice rather than a defect at one site

System B contains zero patients with reference labels. Only 16 of its 1,007 cohort patients have the variables the reference model requires, so no site-native reference assignment exists to compare against. The consequence bounds every transfer claim here: **external transfer could be evaluated operationally and prognostically, but not by direct label agreement.** Scoreability, predicted prevalence and outcome association are measurable; external κ, sensitivity, specificity and calibration against a gold assignment are not.

Predicted prevalence rose from 27.9 % at A to 35.2 % at B. The predicted class continued to stratify sepsis onset in the same direction (22.9 % versus 18.0 %), but **not significantly** ($p = 0.10$, overlapping intervals), where the same comparison at A was decisive ($p < 0.0001$).

3.5 Degradation, and a result that cuts against expectation

Agreement fell from $\kappa = 0.869$ on 12 variables to 0.699 at 7, then to 0.461 at 6 and 0.130 at 2, losing half its full-feature performance by five variables (Figure 3). Performance is comparatively flat from 12 down to 7 and collapses thereafter.

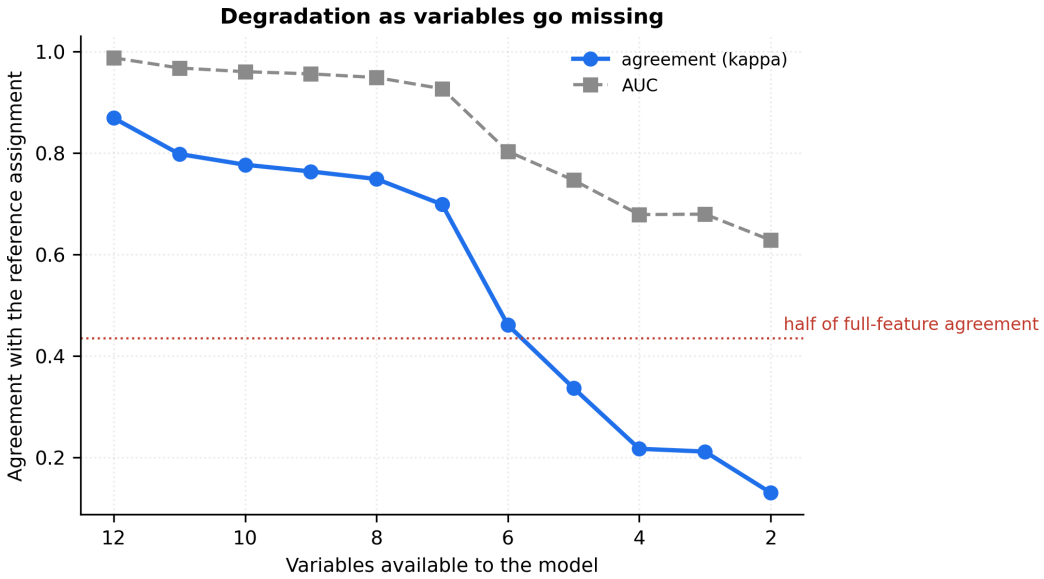


Figure 3. Agreement with the reference assignment as variables are removed least-available-first. The cliff falls between seven and six variables

Natively sparse patients ($n = 2{,}396$, mean 8.5 variables measured) have a **higher** sepsis rate than the well measured (17.1 % versus 14.1 %, $n = 3{,}291$) and are predicted **better** (AUC 0.675 versus 0.592). Sparse data is not degraded data here; it marks a different kind of patient.

The missingness mask alone reached AUC 0.546 against 0.641 for the values, and added nothing when combined with them (0.641 either way). Missingness therefore carries real but modest signal, and the stronger conjecture that it might *beat* the measurements is **not supported**.

3.6 The endotype adds little beyond the available severity score

Predictor	AUC for sepsis onset
Endotype alone	0.544
Modified SOFA alone	0.540
Both	0.556

Table 4. Endotype adds +0.017 AUC over severity, below the 0.02 threshold set before the result was seen

The endotype provided **little incremental discrimination beyond the available severity score**. That phrasing is deliberate: the stronger claim that severity *explains* the classes is not supported, because the comparator is impaired and both predictors are weak. Two qualifications prevent this being over-read. First, **all three values are near chance**: severity does not predict sepsis onset either, so this compares two weak predictors. Second, the severity score is deliberately weakened (Section 2.7), which biases the comparison toward the endotype and makes the null *stronger* than it appears. Classes partially survive severity adjustment: ARI 0.332 between the original assignment and one refitted with features standardised within severity quartile. The confound is real but not total.

3.7 Prognostic stratification replicates in an independent database

Applying the frozen system-A model to the eICU demonstration database — a different schema, 125 hospitals — required mapping laboratory names and one unit conversion, since eICU charts FiO_2 as a percentage where Challenge 2019 uses a fraction. All 14 transferable features were present and passed range verification.

Predicted class	n	Deaths	Hospital mortality	95 % CI
Hypoinflammatory-like	244	41	16.8 %	12.3–22.1 %
Hyperinflammatory-like	142	52	36.6 %	28.7–45.1 %

Table 5. Mortality by predicted class in eICU. Fisher $p < 0.0001$, non-overlapping intervals, a 2.2-fold ratio. What replicates is the mortality *gradient*, not the underlying biological endotype

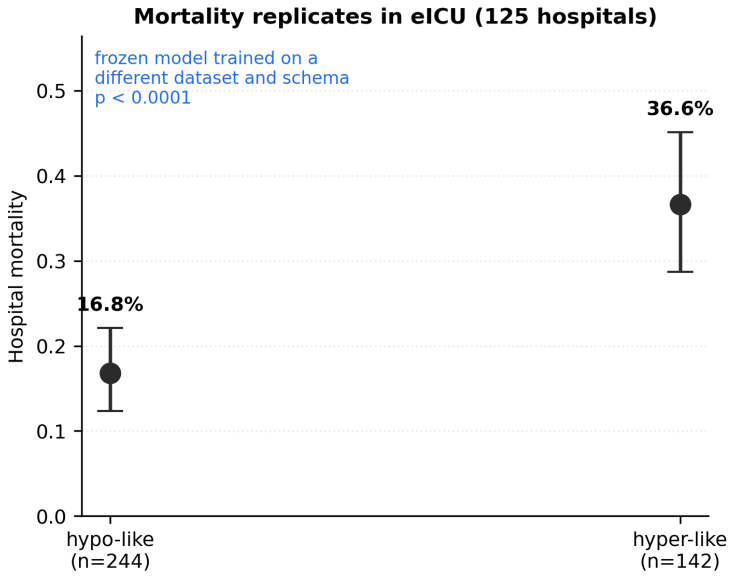


Figure 4. Hospital mortality by predicted class, 95 % intervals. The model was trained on Challenge 2019 system A and frozen; eICU is a different dataset with a different schema

Benchmarked against APACHE IVa — a published, externally validated model whose predictions ship with the data and which was not fitted here — the endotype probability reached AUC 0.634 against APACHE's 0.700. APACHE wins, as expected: it is purpose-built for mortality prediction and an endotype label is not.

3.8 Enrichment: the binding constraint is screening, not accuracy

Unstratified enrollment requires 2,196 patients per arm; perfect enrichment on the high-event class requires 1,573, a 28.4 % reduction; at realistic classifier accuracy, 1,798. **The trial size quoted below is** $3{,}596 = 2 \times 1{,}798$, both arms at that accuracy.

Screening burden is where this becomes consequential. Patients must both test positive and be scoreable at all, so the number screened is $2n/(f_{\text{pos}} \cdot f_{\text{scoreable}})$, where $f_{\text{pos}} = 0.349$ is the fraction testing positive. To enrol the same 3,596 patients at the second hospital, the coverage-matched

model requires screening 12,494 patients and the blood-gas-dependent model 647,654 — a 52-fold difference from an identical trial design (Figure 5).

This ratio is more robust than it may appear. Because trial size, accuracy, prevalence and f_{pos} are identical in both rows, they cancel exactly, and the ratio reduces to the scoreability ratio alone: $0.824/0.0159 = 51.8$. It therefore does not depend on the assumed 20 % relative risk reduction, the 80 % power, or the assumed classifier accuracy — none of which enter the comparison. What it does depend on is the observed coverage difference between the two hospitals.

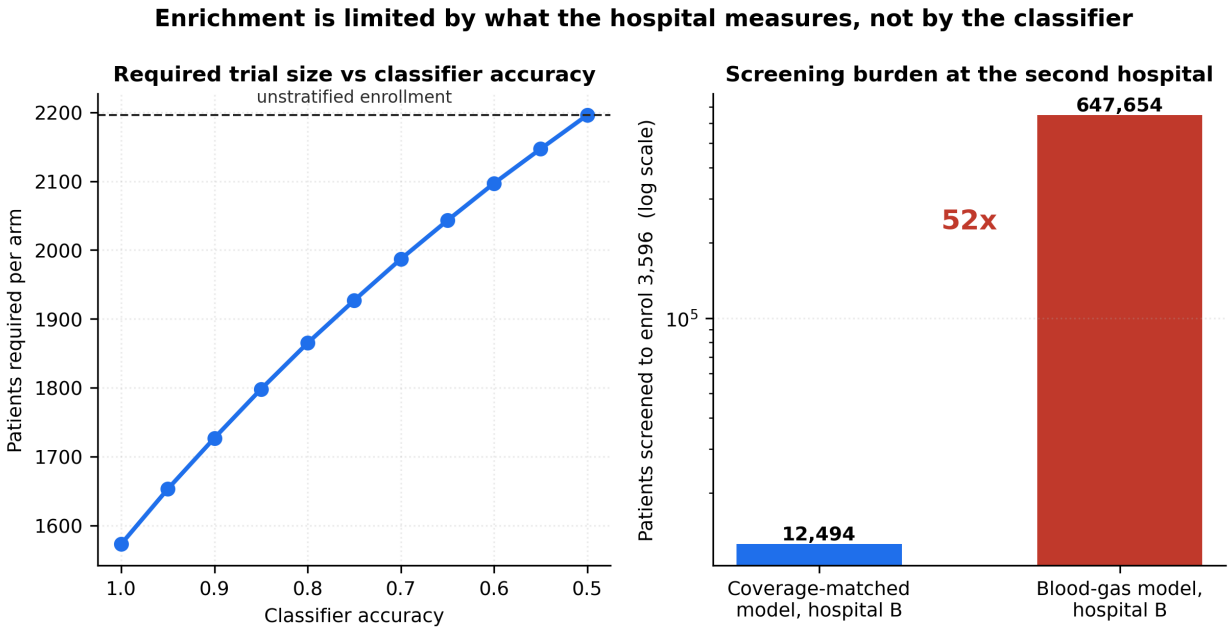


Figure 5. Left: required patients per arm as classifier accuracy falls, against unstratified enrollment. Right: patients who must be screened to enrol the same trial at the second hospital, coverage-matched versus blood-gas variables, log scale

3.9 Refinement

The round-1 error taxonomy identified two concrete failure modes: the model is **inverted** on sparsely measured patients (risk ratio 0.12 versus 1.71 on well-measured), and the signal is carried almost entirely by patients aged 65 and over (1.78 versus 0.98).

Two pre-specified rules fired: threshold recalibration, and the addition of measured-variable count as an explicit feature. On the holdout, scored once, enrichment improved (risk ratio 1.62 to 2.02, crossing into significance at $p = 0.039$) while discrimination did not (AUC 0.634 to 0.629). This is the expected shape rather than a contradiction, since both refinements act on the operating point and AUC is threshold-invariant. **The deployable quantity improved; the model did not.**

3.10 Molecular layer (secondary triangulation)

The MARS endotype labels were present in GSE65682 metadata for 479 samples. A regularised multinomial model predicted them at 87.5 % out-of-fold accuracy with per-class AUC 0.975 to 0.988. Consensus clustering, without access to those labels, partially recovered them (ARI 0.385) — but

PAC remained at or above 0.42 at every k searched, so **no resolution gives a stable partition**, and the selected k again sat at the range boundary.

The frozen signature scored on two held-out cohorts separated sepsis-with-ARDS from sepsis alone in the expected direction but not significantly (Cliff's $\delta = +0.303$, $p = 0.051$, $n = 29$ versus 28). In the longitudinal cohort, sepsis patients' scores rose toward the ARDS level between day 0 and day 7 ($p = 0.024$) while ARDS patients started high and stayed flat.

In Olink proteomics, day-0 class predicted acuity at every subsequent timepoint with the effect **growing** over time (Cliff's δ from -0.344 at day 0 to -0.505 on maximum acuity, all $p \leq 10^{-6}$). **40.3 % of patients changed inflammatory status between day 0 and day 7**, indicating states rather than fixed traits.

The p38 axis separated only marginally better than a disjoint generic-inflammation comparator (Cliff's δ 0.676 versus 0.611, margin 0.065). An interferon negative control showed no separation ($\delta = 0.002$, $p = 0.98$), confirming the comparison detects pathway-specific signal rather than general activation. Per the pre-committed rule, this is **consistent with, but not evidence of, p38-axis dominance**. Clusters did not correspond to COVID status as a partition (ARI 0.053) but were not independent of it either (χ^2, $p = 7 \times 10^{-18}$); the honest reading is partial host-response structure with an aetiology component.

4 Discussion

4.1 What each claim does and does not establish

Claim	Evidence	What it does **not** establish
A two-class directional pattern is recoverable	GMM; four published directions at 28.5 % prevalence	That this partition equals the published latent classes
A portable classifier reproduces the reference assignment	$\kappa = 0.852$; both portability checks passed	External classification accuracy at another site
The model transfers operationally	82.4 % scoreable at system B	External label agreement, calibration, sensitivity/specificity
Measurement practice limits deployment	82.4 % vs 1.6 % scoreable; bicarbonate 97.4 % vs 2.1 %	Why each institution orders what it orders
The classes retain prognostic signal	eICU 16.8 % vs 36.6 % mortality, $p < 0.0001$	Biological endotype replication; treatment-effect heterogeneity
Refinement improved enrichment	RR 1.62 $\rightarrow$ 2.02, holdout scored once	Improved discrimination (AUC was flat)

Table 6. Claims, their evidence, and their bounds

The endpoint, not the classes, was the limiting factor. On the primary cohort the subphenotypes looked close to worthless: AUC 0.544 for sepsis onset, adding +0.017 over a severity score. On eICU the same frozen model separated mortality by a factor of 2.2 with $p < 0.0001$. Sepsis onset is a six-hour-shifted Sepsis-3 label these subphenotypes were never derived against; mortality is what they were built on. Reporting either result without the other would mislead, and the pairing is the single most important interpretive point in this study.

Deployability is governed by institutional measurement practice. Not by documentation quality: a variable absent from a record may never have been *ordered*, which is a different and larger claim. The bidirectional difference supports the broader reading — system B records diastolic pressure and troponin more often, not less — so what differs is which tests each institution routinely obtains. The classifier is not the bottleneck. A model reproducing its reference assignment

at $\kappa = 0.852$ becomes unusable at a hospital that does not order the tests it needs, and the parsimony curve makes this worse rather than better: the most informative small variable set is disproportionately built from blood-gas measurements, which are exactly the measurements whose ordering varies most between institutions. Parsimony and portability pull in opposite directions.

For trial design, this reframes the enrichment question. The literature's attention to classifier accuracy is reasonable but secondary; between a perfect classifier and a realistic one the required trial size moves by a few hundred patients, whereas between two real hospitals the screening burden moves by a factor of 52. A trial-feasibility assessment that does not audit site-level variable coverage is measuring the wrong thing.

On the sparse-patient result. That patients with fewer measured variables have more events and are predicted better is counterintuitive but not inexplicable: measurement intensity encodes clinical trajectory, and the sparsely measured group likely mixes the rapidly deteriorating with the briefly admitted. It is reported here as an observation requiring explanation rather than as an explained finding.

5 Limitations

- **The cohort is not ARDS.** No imaging, no PEEP, no ventilation flag; the definition rests on an oximetry surrogate in an unadjudicated population.
- **The primary cohort has no mortality outcome.** Mortality rests entirely on 386 eICU patients with 93 deaths — thin, with intervals reported wide.
- **The severity comparison is weakened by design**, missing vasopressors and GCS with bilirubin absent for 77 %. This biases toward the endotype, so the null is stronger than it appears, but it is not a like-for-like comparison with published SOFA analyses.
- **The published biomarker classifier cannot be evaluated anywhere in this data.** IL-6 and sTNFR1 are absent from EHR data entirely; Olink NPX is a relative log2 scale rather than a concentration, so a fixed-coefficient equation cannot consume it; and the proteomic cohort's clinical file contains no bicarbonate. The model is implemented correctly and shown to be inapplicable, which is itself the finding.
- k **selection hit the pre-registered range boundary in both layers**, so the range rather than the criterion determined the number of classes in each case. This should not be read as establishing a four-class clinical solution or a six-class molecular one: BIC and PAC were both still improving at the edge of the searched range, so the preferred k is unknown and may be larger. All substantive results use the two-class comparison, chosen for comparability with the published dichotomy rather than by the selection criterion.
- **PAC never fell below 0.42** in the transcriptomic layer: no k gives a stable partition, and the ARI against published labels is doing the work stability cannot.
- The cohort is selected by its own inclusion criteria, and the reference-model subset is selected again by completeness — a selection on a selection.
- The proteomic cohort is COVID-era, so its inflammatory biology is related to but not identical with non-COVID acute respiratory failure.

6 Methodological corrections

Six analyses in this study produced plausible-looking results that were wrong. Each was caught by a test, an assertion or a range check rather than by inspection, and each is recorded because the failure modes are more instructive than the successes.

An ablation that had conditioned away the thing it measured. The first missingness ablation reported every feature at 100 % availability and a mask-only AUC of 0.499 — which reads as "missingness carries no signal." It does not. Endotype labels exist only for the reference model's complete cases, so within the labelled subset the mask is all-ones by construction and a model fitted on it cannot beat chance. Caught by every availability printing as exactly 1.000. *Any analysis of missingness conducted on complete cases is answering a different question than it appears to.*

A comparator that overlapped the thing it was comparing against. The p38 protein set shared 7 of its 10 assays with the "generic inflammation" comparator, so the finding that p38 separated better was substantially the set being compared with itself. A circular comparison producing a *positive* result is the most dangerous kind, because nothing looks wrong. The comparator is now disjoint by construction.

One statistic reported where two disagreed. The host-response result was initially reported from ARI alone (0.053) while a chi-square on the same table gave $p = 7 \times 10^{-18}$. These answer different questions — whether the partitions match, versus whether any dependence exists — and a low ARI with a tiny p means the clusters are not a relabelling of COVID status but some are enriched. The analysis now requires both measures to agree before making the clean claim.

A substring match that swallowed the control group. In GSE66890 the exclusion label literally contains the phrase "sepsis with ARDS" — the full string reads "not meeting the Berlin definition ... and that they were excluded from analysis" — so a substring match folded five explicitly excluded patients into the ARDS arm, inflating it from 29 to 34 and moving the effect size. Now matched exactly.

A time index that was not the index it appeared to be. An ingest checkpoint fired on 8,793 of 40,336 patients. ICULOS is perfectly contiguous for every patient but 22 % *start* after hour 1, because their record begins some hours after admission. ICULOS is hours-since-admission; row position is not. The practical impact was two patients out of 5,689 — recorded anyway, because it is precisely the class of error that shifts a variable amount per patient and never raises anything. The checkpoint that caught it was written before the data was seen.

A clustering that was degenerate rather than unfavourable. The first consensus clustering used Euclidean distance with average linkage and returned one 478-patient cluster plus singletons, with ARI 0.0004. The method was changed to a correlation distance, standard for expression data. **The distinction matters:** the trigger was structural degeneracy — singleton clusters, which the pre-specified criteria already disqualify — not an unfavourable agreement statistic. Changing method because a result is disliked is result-chasing; changing it because the output is broken is diagnosis.

Two traps caught before they entered a result. eICU charts FiO_2 as a percentage where Challenge 2019 uses a fraction; missing this would have made every S/F ratio a hundredfold too large and produced an empty cohort — a failure that would have looked like "the cohort does not exist here" rather than a units bug. And the reference model's sTNFR1 coefficient in common circulation (1.1913) differs from the value in the primary sources (1.1903); the difference is roughly 0.009 log-odds, flips no classification, and would therefore have remained invisible indefinitely.

7 What was not attempted, and why

Predictive enrichment. Which patients *benefit* more from treatment requires a treatment-effect estimate and is confounded by indication in observational data. A causal forest here would return a confident, wrong and dangerous-sounding answer. The appropriate design is a target-trial emulation with negative-control outcomes and E-value sensitivity analysis.

Drug–disease signature mapping. Out of scope, and it would require a molecular layer stronger than the one obtained.

Refitting the reference classifier on NPX. Legitimate as an exercise, but the resulting model would be ours rather than an external reference, and the entire design depends on the reference being external.

A Cohort attrition

Step	System A	System B	Total
All ICU patients	20,336	20,000	40,336
Age $\geq$ 18	20,336	19,978	40,314
$\geq$ 24 h of record	15,837	14,809	30,646
$\geq$ 1 valid S/F candidate hour	6,470	2,078	8,548
S/F $\leq$ 235 (final cohort)	4,680	1,007	5,687

Table 7. CONSORT-style attrition by hospital system

B Reproducing this work

```
conda env create -f environment.yml && conda activate ards
# see DOWNLOADS.md; data acquisition is manual by design
make test     # 33 tests, including the integrity guards
make scope    # descriptive scoping
```

No data is committed. Every deviation from the original analysis plan is recorded in `DECISIONS.md` with the observation that triggered it.

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